

# **TECH STACK & DETAILED PRACTICAL USAGE**

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## 1. Development Environment & Coding Tools

I primarily use Visual Studio Code as my main development environment because of its flexibility, speed, and strong extension ecosystem. It allows me to work efficiently across different types of projects, including web development and basic AI experiments. I use extensions like Live Server for real-time previews and Prettier to maintain consistent formatting, which improves readability and code quality.

In addition to coding, I rely on Git and GitHub for version control. These tools help me track changes, manage different versions of my code, and maintain a structured workflow. They are especially useful when working on projects over time, as they allow me to revisit previous versions and understand how my code has evolved.

A practical example of this is my React To-Do List project. I used Visual Studio Code to write and debug code, Git to commit changes after each feature implementation, and GitHub to store and manage the project repository. This experience helped me understand how professional development workflows are structured.

## 2. Web Frameworks

I use React.js as my primary web framework to build interactive and dynamic user interfaces. React enables me to create reusable components, which makes my code more organized and scalable. It also helps in managing application state efficiently, which is important for responsive user experiences.

In my projects, I focus on structuring applications into smaller components, each responsible for a specific functionality. This modular approach makes it easier to debug, maintain, and extend the application over time.

For example, in my To-Do List application, I created separate components for input handling and task display. I used state management to update tasks dynamically when users added or removed items. This helped me understand how real-time updates work in frontend applications.

## 3. AI/ML Frameworks

I have explored machine learning frameworks like TensorFlow and scikit-learn to understand how models are built and trained. These frameworks provide tools for handling data, training models, and evaluating performance.

My approach has mainly been experimental, where I focus on learning core concepts such as data preprocessing, model training, and evaluation metrics. This has helped me build a foundational understanding of how machine learning systems operate.

For instance, I used scikit-learn to build a simple classification model. I prepared the dataset, trained the model, and evaluated its accuracy. This hands-on experience helped me understand the complete workflow of machine learning.

## 4. Agentic AI Tools

I have experimented with tools like LangChain and AutoGPT to understand how AI agents perform tasks autonomously. These tools are designed to break down complex problems into smaller steps and execute them

sequentially.

My exploration involved understanding how prompts can be chained together to create workflows that simulate reasoning and decision-making processes.

In one experiment, I used LangChain to process input through multiple stages and generate structured output. This helped me understand how AI systems can automate multi-step tasks.

## **5. LLM Platforms**

I frequently use ChatGPT as a learning and development assistant. It helps me understand new concepts, debug code, and explore different approaches to solving problems.

It plays an important role in my workflow by providing explanations and suggestions that improve my understanding and efficiency.

For example, while working on programming and DBMS-related problems, I used ChatGPT to get step-by-step explanations and improve my problem-solving skills.

## **6. AI Tools**

I use GitHub Copilot to enhance my coding productivity by providing real-time code suggestions. It helps me write code faster and reduces the chances of syntax errors.

This tool is especially useful when working on repetitive code patterns, as it suggests optimized implementations.

In my React projects, Copilot helped me write event handling logic and state updates more efficiently, which improved both speed and code quality.

## **7. Cloud Platforms**

I have experience using Firebase as a cloud platform for deploying applications and managing backend services. It simplifies the process of hosting and provides built-in features like authentication and database integration.

My usage includes deploying small projects and understanding how cloud services handle application data and user interactions.

For example, I used Firebase to host a web application and explored authentication features, which helped me understand real-world deployment processes.

## **8. Databases**

I work with both MySQL and MongoDB to handle different types of data. MySQL is used for structured data and supports SQL queries, while MongoDB is useful for flexible, JSON-based data storage.

My experience includes writing queries, designing schemas, and performing data operations, which are essential for backend development.

In my coursework and small projects, I used MySQL to manage structured data and MongoDB to perform CRUD operations, helping me understand how different database systems are used in real-world applications.